

Firearm Purchase Demand Analysis

US State-Level Purchase Demand Analysis — 2019 to 2023

Rajarajeswari Sankar

Item	Detail
Dataset	FBI NICS Firearm Background Checks — Nov 1998 to Sep 2023
Scope	51 jurisdictions (50 states + DC) 16,445 rows 27 columns
Metric	demand_checks = handgun + long_gun + multiple + other + permit
Dashboard	https://rajarajeswari-sankar-data-science-2026.shinyapps.io/nics-firearm/
Objective	Analyse how firearm purchase demand varies across US states over time

1 — Dataset and Preparation

1.1 Dataset Overview

The FBI National Instant Criminal Background Check System (NICS) dataset contains monthly counts of background checks by type for each US state from November 1998 to September 2023. Each row represents one state in one month across 27 columns covering 24 distinct check-type categories and a monthly total column. The 24 check types span direct purchase checks, administrative processes, pawn transactions, private transfers, rentals, and returns.

1.2 Data Preparation

- Month column converted to consistent date format and records arranged chronologically by state
- Four US territories removed — Guam, Mariana Islands, Puerto Rico, and Virgin Islands — due to inconsistent reporting structures differing from US states
- District of Columbia retained as it reports consistently and provides a useful policy contrast given its highly restrictive gun laws
- Analysis filtered to 2019–2023 for state-level demand analysis; full 1998–2023 period used for national trend context in supplementary figures
- Rows with invalid growth values arising from zero baseline months excluded using conditional filtering

2 — Demand Metric Construction

2.1 Why Not Raw Totals?

Raw NICS totals include administrative permit rechecks, pawn transactions, and range rentals that do not represent new firearm purchases. Using raw totals would artificially inflate counts in states like Illinois and Kentucky — which run periodic administrative permit rechecks — making cross-state comparison invalid. The FBI explicitly cautions against using raw totals for cross-state comparisons for this reason.

2.2 Demand Checks Formula

$$\text{demand_checks} = \text{handgun} + \text{long_gun} + \text{multiple} + \text{other} + \text{permit}$$

These five check types represent background checks initiated for new firearm ownership. Permit checks are included because some states such as Illinois, Hawaii, and Alabama route purchases through a permit system rather than direct point-of-sale checks. In these states, permit and handgun checks are functional substitutes — the purchase still occurs but is counted at the permit stage rather than at the counter (Nemerov, 2018).

2.3 Excluded Columns

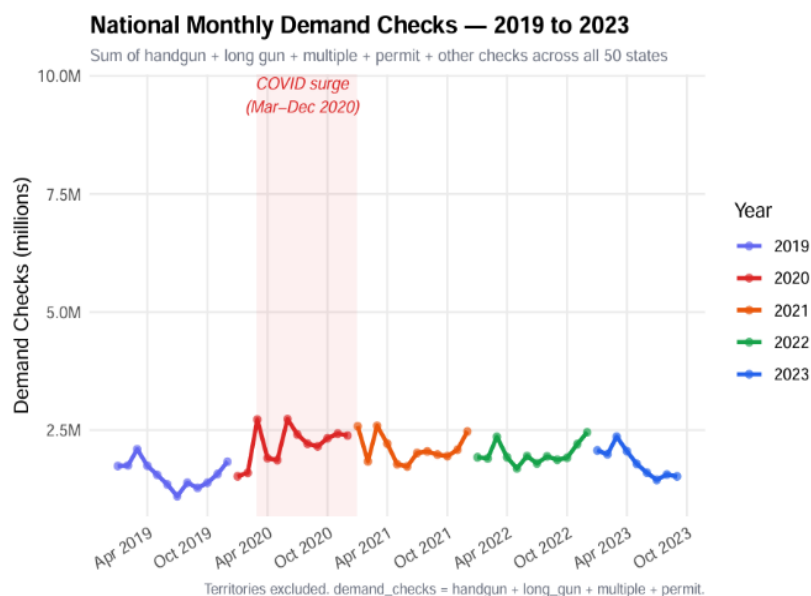
Column	Reason for Exclusion
permit_recheck	Administrative recheck of existing permits — no new purchase occurs
admin	System-level check — no firearm transaction involved
prepawn_*	Person pawning a gun they already own — gun leaving not entering possession
redemption_*	Reclaiming a previously pawned gun — not a new purchase
returned_*	Gun returned to dealer — opposite direction of a purchase
return_to_seller_*	Same as returned — gun going back not forward into new ownership
rentals_*	Range rental only — no ownership transfer whatsoever
private_sale_*	Inconsistently reported across states — cross-state comparison invalid

The unifying principle: only check types where a new firearm enters someone's possession for the first time through a purchase transaction are included.

3 — Analysis and Findings

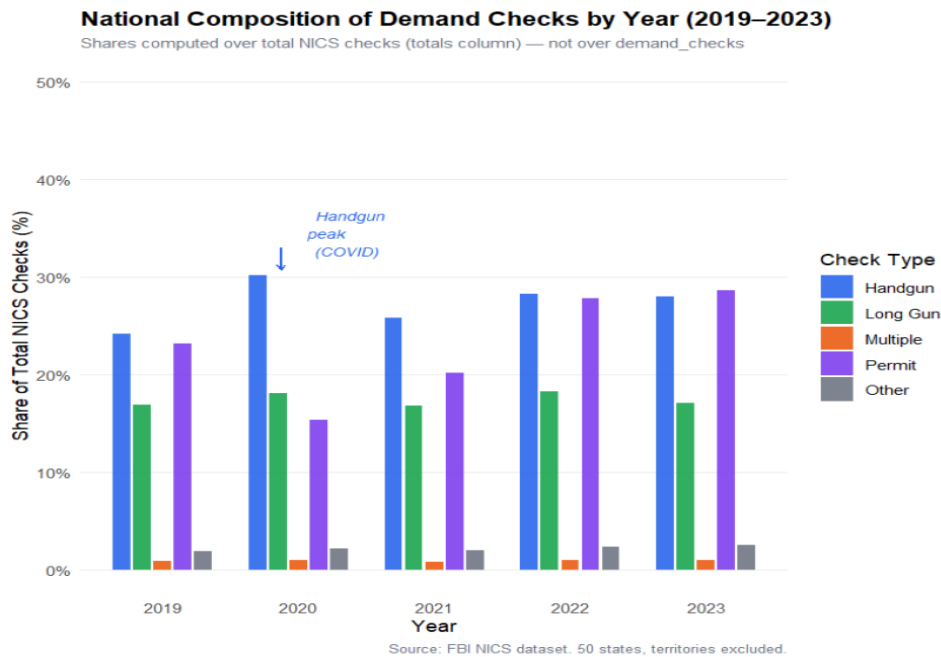
3.1 National Monthly Demand (2019–2023)

National monthly demand checks aggregated across all 50 states and DC from 2019 to 2023 establishes the baseline context for all state-level analysis. In 2019 demand was stable at around 1.5 to 2 million checks per month nationally — the pre-pandemic equilibrium. Beginning March 2020, demand surged sharply to nearly 3.5 million checks in a single month, more than double the 2019 baseline, driven by COVID-19 pandemic uncertainty and unprecedented first-time buyer demand. Through 2021 demand remained elevated but began declining from the 2020 peak. By 2022 and 2023 the trend returned toward pre-pandemic levels with seasonal patterns re-emerging, indicating the COVID surge was a temporary shock rather than a structural shift in baseline demand.



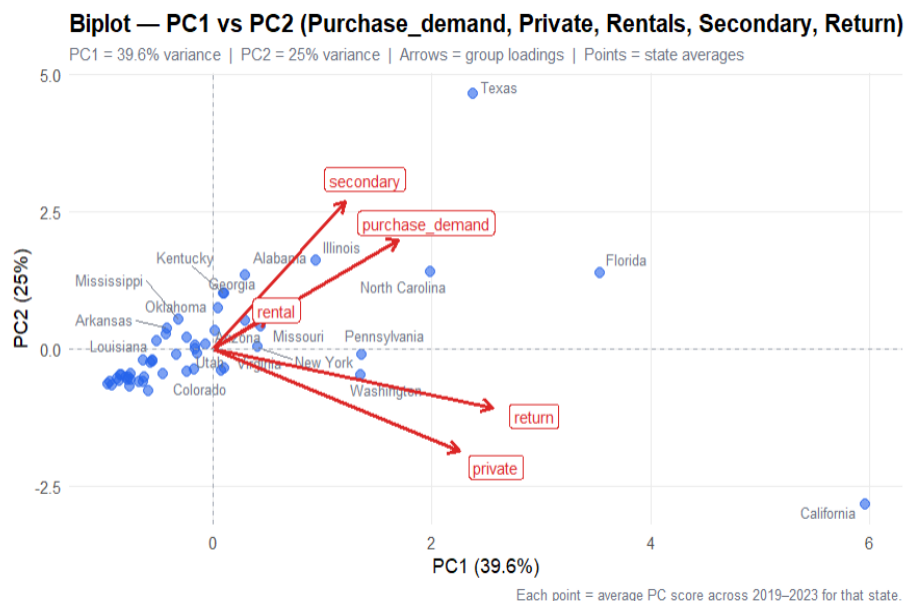
3.2 National Composition of Demand Checks (2019–2023)

The composition chart shows how the mix of check types shifted at the national level across the five years. Handgun share peaked at approximately 31% in 2020 — the highest across all five years — consistent with COVID-driven first-time buyer demand for home protection. Long gun share remained broadly stable at 17 to 19% across all years, reflecting existing-owner behaviour rather than reactive new-buyer demand. Permit share is structurally stable at 20 to 29% throughout — this reflects states that permanently route purchases through the permit system. It does not fluctuate with demand shocks because it is a policy feature not a consumer behaviour signal. Multiple and other combined remain consistently below 3%, confirming they are minor volume contributors but are correctly included as direct purchase signals.



3.3 PCA Analysis — Structural Validation

To understand the structural dimensions of inter-state variation, Principal Component Analysis was conducted on five conceptual groups derived from the 24 check-type columns (Figure 4). PC1 captures 39.6% of total variance and is dominated by purchase demand, confirming it is the primary axis separating states from each other. Secondary market activity loads in a similar direction to purchase demand on PC1, as both are driven by overall state market volume. Private transfers and returns load on PC2 (25% variance), representing a structurally separate dimension of state behaviour. California and Texas appear as high-volume outliers on PC1 due to their large populations. The biplot confirms that purchase demand is the dominant differentiating factor across states, statistically validating the demand_checks metric construction.



4 — Interactive Dashboard

The interactive Shiny dashboard presents state-level purchase demand analysis through three coordinated panels controlled by a year dropdown (2019–2023). Clicking any state on either map updates the category breakdown bar chart for that state and year. The dashboard is accessible at: <https://rajarajeswari-sankar-data-science-2026.shinyapps.io/nics-firearm/>

4.1 Average Monthly Demand Growth Rate

The growth rate map shows how firearm demand varies across states each year (Figure 2). The RdBu scale reflects blue for increase in demand, red for decline, and white for little change. Clear variation between states is visible in 2019, with Mississippi showing the fastest growth due to anticipatory buying ahead of state elections. In 2020 most states show only mild blue values between 0.05 and 0.10 because averaging monthly growth across all twelve months dilutes the sharp demand spike of early 2020, making the COVID surge less visible on this map than it was in reality. After the 2020 surge many states show declining growth as demand corrected from the elevated peak. State differences reappeared in 2022, with Oregon standing out due to anticipatory buying ahead of Measure 114 gun legislation (Bollman et al., 2025). By 2023 most states show near-zero or negative growth, indicating the market is stabilising.

4.2 Demand Trend Within Year — Regression Slope

The slope map captures whether demand is rising or falling consistently over the course of each year (Figure 2). Reading both maps together provides a complete picture — growth shows the average level of change while slope shows the direction and momentum. On the slope map darker purple indicates negative slopes, yellow indicates strongly positive slopes, and mid-green indicates near-zero slope. In 2019 most states appear similar in green shades while Kentucky stands out with a negative slope. In 2020 the slope map reveals differences not visible in the growth map — Pennsylvania shows the strongest positive slope, with Florida and Michigan also showing rising patterns, meaning demand kept accelerating throughout 2020 rather than just spiking early. Nearly all states show a negative trend in 2021, whereas Illinois stands out as the only state where demand maintained a consistently positive monthly trend. In 2022 Illinois continues to show the highest positive slope for a second consecutive year, confirming a structural pattern rather than a single-year anomaly. By 2023 most states show green shading indicating near-zero slope, while Illinois sharply reverses to the most negative slope in the country after two years of leading positive momentum. Together the two maps provide complementary information: growth is the scale of change and slope is the direction. High growth with a negative slope implies a temporary spike followed by correction, while high growth with a positive slope implies sustained demand increase.

4.3 Category Composition Bar Chart

The bar chart shows the structural composition of firearm demand in each state, breaking down the proportion of each check type that drives demand and helping explain the patterns visible on the maps (Figure 2). Clicking Michigan in 2020 reveals that handgun checks have the highest proportion, consistent with COVID-driven first-time buyer demand for personal

protection. The permit share in Illinois is relatively large — exceeding 80% in 2021 and 2022 — which directly explains why Illinois stands out on the slope map in those years. The high permit share reflects Illinois's administrative routing of purchases through its permit system rather than a genuine surge in new purchase demand. Firearm demand is highly sensitive to major events such as elections, COVID, and policy changes (Figure 3). Raw NICS totals include permit rechecks and other administrative checks that do not accurately represent purchases. Focusing on direct purchase checks shows that handgun checks dominate in most states, while high permit shares reflect state administrative procedures rather than genuine demand differences.

5 — Summary of Findings

Year	Growth Map Finding	Slope Map Finding	Key State
2019	Clear geographic variation	Most states flat near-zero	Mississippi — highest growth Kentucky — negative slope
2020	Mild uniform blue (0.05–0.10) COVID surge diluted by averaging	Pennsylvania strongest positive Florida and Michigan rising	Pennsylvania — sustained growth
2021	Near-universal negative Post-COVID correction	Near-universal negative	Illinois — only positive slope
2022	Geographic variation returns	Illinois positive second year	Oregon — Measure 114 surge
2023	Near-zero stabilisation	Illinois sharp reversal negative	Market stabilising nationally

Key Conclusions

- Firearm purchase demand varies substantially across states and is highly sensitive to policy events, elections, and external shocks
- The 2020 COVID surge temporarily eliminated inter-state variation — a national shock large enough to override structural differences between states
- Illinois consistently shows permit-dominated composition exceeding 80%, reflecting administrative routing rather than genuine demand variation
- Oregon's 2022 growth spike is directly attributable to anticipatory buying ahead of Measure 114 gun legislation
- PCA confirms purchase demand is the dominant axis of inter-state variation, statistically validating the demand_checks metric
- Reading growth rate and slope maps together is essential — a state can show high growth but declining slope indicating a temporary spike, or sustained growth with positive slope indicating structural demand increase

References

- [1] Nemerov, H.R. (2018). Estimating guns sold by state. Social Science Research Network. doi:10.2139/ssrn.3100289
- [2] Bollman, K.M., Hansen, B., Rubin, E., & Stanford, G. (2025). Gun policy and the Steel Paradox: Evidence from Oregonians. Social Science Research Network. doi:10.2139/ssrn.5103550
- [3] FBI (2024). NICS Firearm Checks: Month/Year by State/Type. Washington: FBI. Available at: www.fbi.gov/services/cjis/nics